

Big Bang Nucleosynthesis (BBN)

Light-element abundance curves H, ⁴He, D, ³He, ⁷Li vs cosmic time / temperature. Weak freeze-out, neutron decay gap, deuterium bottleneck → Y_p ≈ 0.245. Slide η₁₀ and N_{eff} over the classic BBN curves; observed values overlaid.



BBN PARAMETERS

η₁₀ (10¹⁰ × n_b/n_γ) 6.1

N_{eff} (effective neutrinos) 3.046

Planck CMB (η₁₀=6.1, N_{eff}=3.046)

Low η₁₀ = 2 (D high)

High η₁₀ = 12 (D low, Li7 high)

N_{eff} = 4 (one extra ν)

- ☒ show n/p ratio
- ☒ show D / H
- ☒ show ³He / H
- ☒ show ⁷Li / H
- ☒ show observed values (right edge)
- ☒ show top temperature axis (MeV)

Reset to Planck fiducial

SHORTCUTS

- Slide η₁₀ to see how the surviving abundance of D drops as you add more baryons

MEASURED VALUES

Y_p (He-4 mass) 0.2486

D / H 2.532 × 10⁻⁵



About this model

Light-element Big Bang nucleosynthesis (BBN) curves track mass fractions of H, ⁴He, D, ³He, and ⁷Li versus cosmic time and temperature. Weak freeze-out sets the n/p ratio; a neutron-decay gap follows; then the deuterium bottleneck delays fusion until T is low enough for D to survive photodissociation, after which ⁴He locks most neutrons into Y_p ≈ 0.245. Sliders η_{10} (baryon-to-photon ratio ×10¹⁰) and N_{eff} (effective neutrino species) shift the classic abundance curves, with observed bands overlaid. The network is pedagogical: incomplete reaction rates, no full nuclear Monte Carlo, and no stellar astration modeling. Vary η_{10} and N_{eff} to see how helium and deuterium respond — the classic probe of early-universe expansion and baryon density.

Who it's for: Cosmology and nuclear astrophysics courses covering BBN abundances and early-universe parameters.

KEY TERMS

- Big Bang nucleosynthesis
- helium mass fraction
- deuterium bottleneck
- baryon-to-photon ratio
- N_{eff}
- weak freeze-out

How it works

Big-Bang nucleosynthesis (BBN) — within the first ~3 minutes the universe forged the bulk of its primordial light elements. Watch the **n/p ratio** drop from equilibrium toward freeze-out at **T_F ≈ 0.8 MeV**, decay slowly through neutron β-decay, and then suddenly lock into **He-4** when the deuterium bottleneck breaks at **T ≈ 0.07 MeV**. Trace amounts of **D**, **³He** and **⁷Li** survive. The asymptotic abundances depend almost only on the **baryon-to-photon ratio η_{10}** (and a little on the **effective neutrino number N_{eff}**) — moving the slider walks you across the famous BBN curves and explains why measuring just **D/H** in distant quasars or **Y_p** in HII regions pins down Ω_b h² independently of the CMB. Right-edge ticks mark the observationally inferred values; note the long-standing ⁷Li discrepancy.

Key equations

$$T(t) = (1.32 \text{ s} \cdot \text{MeV}^2)^{1/2} / \sqrt{t} \text{ (radiation era)}$$

$$(n/p)_{\text{eq}} = \exp(-Q/T), \quad Q = m_n - m_p = 1.293 \text{ MeV}$$

$$Y_p \approx 2(n/p) / (1 + n/p) \text{ at BBN onset}$$

$$D/H \propto \eta_{10}^{-1.6}, \quad {}^3\text{He}/H \propto \eta_{10}^{-0.6}, \quad {}^7\text{Li}/H : \text{U-shape vs } \eta$$

Frequently asked questions

Why is most helium already set by BBN?

Once deuterium forms past the bottleneck, nearly all available neutrons are quickly assembled into ^4He . That yields $Y_p \approx 0.25$ with only weak dependence on later stellar processing for the primordial floor. Stars make more metals and some helium, but the bulk cosmic ^4He fraction is a Big Bang relic.

What does increasing η_{10} do to deuterium?

Higher baryon density means more efficient burning of D into heavier nuclei, so residual deuterium falls steeply with η_{10} . Helium Y_p rises only mildly. That is why measured primordial D/H is a sharp baryometer. A misconception is that all light elements scale the same way with η — they do not.

How does N_{eff} change the curves?

Extra radiation speeds early expansion, freezing a higher n/p ratio and typically raising Y_p while shifting freeze-out timing. The simulator’s N_{eff} slider shows that expansion-rate lever next to η_{10} . Full neutrino decoupling physics is simplified; treat the curves as standard pedagogical BBN trends with observed values marked for comparison.

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Schematic Ia rise/decay vs II-P plateau; standard-candle note.

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Nuclear Binding Curve

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Qualitative B/A vs A with fusion/fission context.

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Cosmic Microwave Background temperature D_ℓ vs ℓ with Sakharov peaks: tune $\Omega_b h^2, \Omega_c h^2, n_s, A_s, \tau, \dots$

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Pair Production Threshold & $\sigma(E_\gamma)$

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Neutron star TOV: toy M–R (polytrope)

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Pair creation channels $\gamma + \text{nucleus} \rightarrow e^+e^-$ (Bethe–Heitler, $\sigma \propto Z^2$), the higher-threshold triplet $\gamma + e^- \rightarrow \dots$

$n_p n_e / n_H$ vs T and ionized fraction X; steep Boltzmann knee ~3000 K scale (density dependent).

Schwarzschild TOV for polytropes $n = 1$ or $3/2$; K normalized so the ρ_c scan peaks near $\sim 2 M_\odot$...